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EET-203

MEASUREMENTS AND INSTRUMENTATION

ANUVINDA V



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TEXTBOOKS

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3. Kalsi H. S., Electronic Instrumentation, 3/e, Tata McGraw Hill, New Delhi, 2012
4. S Tumanski, Principles of electrical measurement, Taylor & Francis.
5. David A Bell, Electronic Instrumentation and Measurements, 3/e, Oxford



MODULE 1

Measurement standards–Errors-Types of Errors- Statistics of errors, Need for calibration.

Classification of instruments, secondary instruments–indicating, integrating and recording- operating forces - essentials of indicating instruments - deflecting, damping, controlling torques.

Ammeters and voltmeters - moving coil, moving iron, constructional details and operation, principles shunts and multipliers – extension of range.



MEASUREMENTS

- Process by which one can convert physical parameters into meaningful numbers.
- Measurement of a given quantity is an act of comparison between the quantity(whose magnitude is unknown) and a predefined standard.
- The numerical measure is meaningless unless followed by a unit.
- For the result of measurement to be meaningful, there are two basic requirements,
 - 1)the standard used for comparison purposes must be accurately defined and should be commonly accepted.
 - 2)the apparatus used and the method adopted must be provable.



METHODS OF MEASUREMENTS

- Broadly classified into two.
 - 1) Direct method
 - o The unknown quantity is directly compared against a standard
 - o The result is expressed as a numerical number and a unit.
 - 2) Indirect method
 - o Use measurement systems



STANDARDS OF MEASUREMENT

- A standard is defined as something that is set up and established by an authority as rule of the measure of a quantity like length, weight, value or quality.
- For example, a meter is a standard established by an international organization for measurement of length.
- Industry, commerce, international trade in modern civilization would be impossible without a good system of standards.
- The role of standards is to achieve uniform, consistent and repeatable measurements throughout the world.



STANDARDS OF MEASUREMENT

- Depending upon the degree of accuracy required for the work, the standards are divided into following four categories or grades:
 1. Primary Standards (Reference Standards).
 2. Secondary Standards (Calibration Standards)
 3. Tertiary Standards (Inspection Standards).
 4. Working Standards (Workshop Measuring Standards).



STANDARDS OF MEASUREMENT

1. Primary Standards (Master Standards)

- o The primary standard is also known as Master Standard, and is preserved under the most careful conditions.
- o These standards are not commonly in use.
- o They are used only after long intervals.
- o They solely used for comparing the secondary standards.
- o Sometimes it is also called Reference Standards.



STANDARDS OF MEASUREMENT

2.Secondary Standards (Calibration Standards)

- o The secondary standard is more or less similar to the primary standard.
- o They are nearly close in accuracy with primary standards.
- o The secondary standard is compared at regular intervals with primary Standards and records their deviation.
- o These Standards are distributed to a number of places where they are kept under safe custody. They are used occasionally for comparing the territory standards.



STANDARDS OF MEASUREMENT

3. Tertiary Standards (Inspection Standards)

- o The Tertiary standard is the first standard to be used for reference purpose in workshops and laboratories.
- o They are used for comparing the working standards.
- o These are not used as frequently and commonly as the working standards but more frequently than secondary standards.
- o Tertiary standards should also be maintained as a reference for comparison at intervals for working standards.



STANDARDS OF MEASUREMENT

4. Working Standards (Workshop Measuring Standards)

- o The working standard is used for actual measurement in workshop or laboratories by the workers.
- o These standards should also be as accurate as possible to the tertiary standard.
- o But sometimes, lower grades of materials can be used for their manufacturing to reduce cost.



STATIC CHARACTERISTICS OF MEASURING INSTRUMENTS

- Static error
- Accuracy
- Precision
- Linearity
- Sensitivity
- Reproducibility
- Drift
- Dead zone
- Resolution
- Scale range and scale span

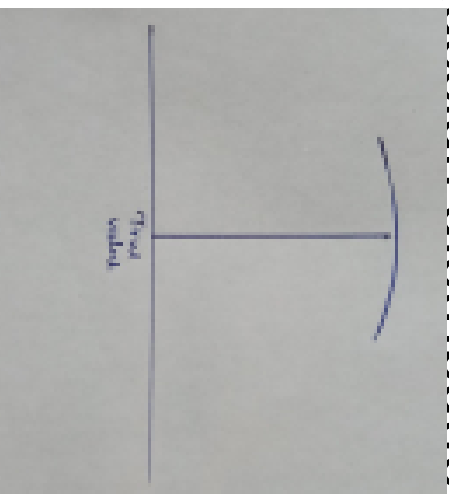
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► True Value of a quantity ????

- o The average of an infinite number of measured values when the average deviation due to the various contributing factors tends to zero.

► While measuring a quantity, we always rely on a set of readings instead of a single reading. The set of readings will be in normal / Gaussian distribution.



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STATIC ERROR

- It is the difference between measured value and true value of the quantity.



δA is error

A_m is measured value of quantity

A_t is true value of quantity

- δA is also called absolute static error (E_o)



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STATIC ERROR

► Relative Static Error (RSE)

- o Ratio of absolute static error (δA) to true value (A_t).

$$RSE = Er = \delta A / A_t$$

$$\% Er = Er * 100$$

► Static Correction

- o Difference between true value and measured value of the quantity.

$$\delta C = A_t - A_m = -\delta A$$



PROBLEMS

- A meter reads 127.5 V and the true value of the voltage is 127.43 V. Determine a) static error, b) static correction for this instrument.

Answer

a)

$$= 127.50 - 127.43$$

$$= +0.07 \text{ V}$$

b) $\delta C = -\delta A$

$$= -0.07 \text{ V}$$



PROBLEMS

- A thermometer reads 95.45°C and the static correction given in the correction curve is -0.08°C . Determine the true value of the temperature.

o Answer

$$\delta C = -0.08^{\circ}\text{C}$$

$$\delta A = +0.08^{\circ}\text{C}$$

$$\begin{aligned} A_t &= 95.45 - 0.08 \\ &= 95.37^{\circ}\text{C} \end{aligned}$$



PROBLEMS

- A voltage has a true value of 1.50V .An analog indicating instrument with a scale range of 0-2.50V shows a voltage of 1.46V . What are the values of absolute error and correction. Express the error as a fraction of the true value and the full scale deflection (fsd).

Answer

- Absolute error $\delta A = A - A$
 $= 1.46 - 1.5$
 $= -0.04V$
- Absolute correction $\delta C = -\delta A$
 $= +0.04 V$
- %RSE = %Er = $(\delta A / A) * 100$
 $= -2.67 \%$
- Relative error (expressed as a % of fsd) = $(-0.04 / 2.5) * 100$
 $= -1.60 \%$



ACCURACY AND PRECISION

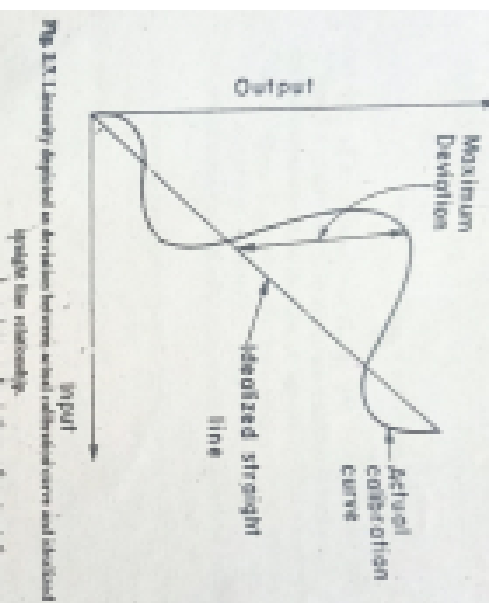
➤ Accuracy and precision same???

- In ordinary usage , the distinction between accuracy and precision is usually very vague.
- But in the field of measurement, they are different.
- Accuracy is the closeness with which an instrument reading approaches the true value of the quantity being measured.
- Precision is the capability of an instrument to repeat and reproduce the same reading.
- Precision does not guarantee accuracy.
- Accuracy leads to precision.



LINEARITY

- An instrument is considered to be linear if its output is linearly proportional to the input.
- Linearity usually reported as non linearity, which is the maximum of the deviation between the calibration curve and a straight line positioned.



SENSITIVITY

- Ratio of change in output to change in input.
- In case of measuring instruments high sensitivity is always preferred for higher accuracy.

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REPRODUCIBILITY AND DRIFT

- Reproducibility is the degree of closeness with which a given value may be repeatedly measured.
- Perfect reproducibility means that the instrument has no drift.
- No drift means that with a given input, the measured value do not vary with time.

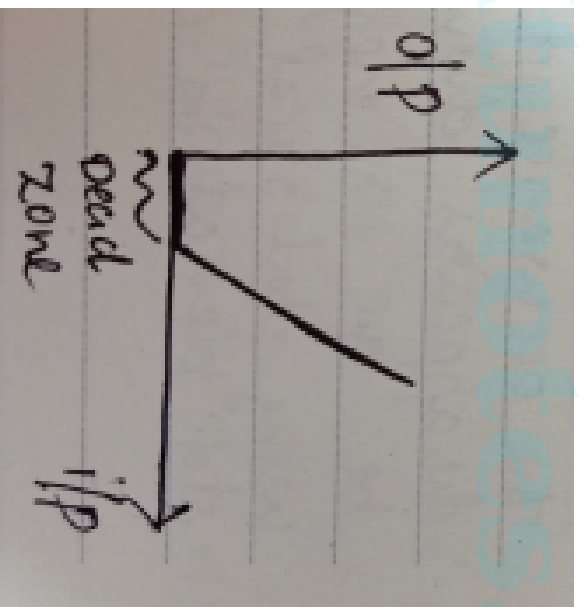


- **Difference b/w repeatability and reproducibility**
 - **Repeatability**-same meter , same observer, conducting experiments under same conditions
 - **Reproducibility**-same meter , different observer and conducting experiment under same conditions.



DEAD ZONE

- This is the minimum value of input beyond which response will occur in an instrument.
- It is also called as threshold.



RESOLUTION

- It is the smallest output we can detect in the scale of an instrument with certainty and clarity.
- Resolution= Full scale deflection/Total no of divisions
- As the number of divisions increases, resolution increases.

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SCALE RANGE AND SCALE SPAN

- Scale range is the largest reading that an instrument can read.
- Scale span is the difference between the largest and the smallest reading of the instrument.
- For example, the largest value an instrument can read is 'y' units and smallest value it can read is 'x' units.
 - Then scale range is y (or x to y)
 - -scale span is y-x



ERRORS IN MEASUREMENT

➤ Limiting errors (Guarantee errors)

- In order to assure the purchaser of the quality of the instrument, the manufacturer guarantees certain accuracy.
- The manufacturer has to specify the deviations from the nominal value of a particular quantity.
- The limits of these deviations from the specified value are defined as limiting errors or guarantee errors.



ERRORS IN MEASUREMENT

- If the magnitude of a quantity has a nominal value of 'As' and maximum error or limiting error of $\pm\delta A$ must have a magnitude 'Aa' between the limits As - δA and As + δA
- Actual value of quantity $Aa = As \pm \delta A$
- For example nominal magnitude of a resistor is 100Ω with a limiting error of $\pm 10\Omega$, then the magnitude of resistance between the limits;

$$Aa = 100 \pm 10\Omega \quad \text{or} \quad Aa \geq 90\Omega \quad \text{and} \quad Aa \leq 110\Omega$$



ERRORS IN MEASUREMENT

- Relative or fractional limiting error
 - It is the ratio of error to the specified(nominal) magnitude of a quantity.

$$\text{Relative limiting error } \epsilon_r = \frac{\Delta A}{A_s} = \frac{\epsilon_0}{A_s}$$

$$\epsilon_0 = \delta A = \epsilon_r A_s$$

$$\epsilon_r = \frac{\epsilon_0}{A_s} = \frac{\delta A}{A_s} = \frac{\epsilon_r A_s}{A_s}$$

$$\% \text{ limiting error} = \% \epsilon_r = \epsilon_r \times 100$$



PROBLEM

- The value of capacitance of a capacitor is specified as $1\text{ }\mu\text{F} \pm 5\%$ by the manufacturer. Find the limits between which the value of the capacitance is guaranteed.

Solution

$$A_a = A_s (1 \pm \epsilon_r) \\ = 1 \times (1 \pm 0.05) = 0.95 \text{ to } 1.05\text{ }\mu\text{F}$$



TYPES OF ERRORS

- Errors may originate in a variety of ways, and are usually classified into three main categories
 - (1) Gross errors
 - (2) Systematic errors
 - (3) Random errors(Accidental errors)



TYPES OF ERRORS

➤ Gross Errors:

- Gross errors are due to human mistakes in reading or using instruments & in recording and calculating measurement results .
- Complete elimination is not possible
- For example, person may read a pressure gauge indicating 1.01 N/m^2 as 1.10 N/m^2 .
- Gross error may be of any amount and therefore their mathematical analysis is impossible.
- However they can be avoided by adopting two means,
 - 1) Great care should be taken in reading and recording the data
 - 2) Two, three or even more readings should be taken for the quantity under measurement, preferably by different experimenters



TYPES OF ERRORS

► Systematic errors

- Systematic errors are divided into three categories
 - i) Instrumental errors
 - ii) Environmental errors
 - iii) Observational errors



TYPES OF ERRORS

i) Instrumental errors

- Instrumental errors arise due to inherent shortcomings in the instrument, misuse of the instrument and due to loading effects of instrument
- **Errors due to inherent shortcomings of instrument** refers to the errors inherent in the instruments because of their mechanical structure
- They may be due to construction, calibration or operation of the instruments.
- This cause the instrument to read too low or too high.
- An example is, if the spring of a permanent magnet instrument become weak, the instrument will always read high.



TYPES OF ERRORS

- Can be avoided by (1) selecting a proper measuring device for the particular application, (2) applying correction factors after determining the magnitude of instrumental errors & (3) careful recalibration of the instrument.
- **Misuse of instruments** /using instruments in an unintelligent way also may give erroneous results.
 - Examples of this misuse of instrument may be failure to adjust the zero of the instrument, poor initial adjustments and using leads of too high resistance etc.
- **Errors due to loading effect** occur due to the use of an instrument with lower sensitivity with heavier load.



TYPES OF ERRORS

ii) Environmental errors

- This error is due to the use of an instrument in different conditions than in which it was assembled & calibrated.
- These may be due to change in temperature, humidity, altitude, earth's magnetic field, gravity, stray electrostatic & magnetic Fields .
- Can be reduced by taking the following precautions.
 - a) Using instrument in controlled conditions.
 - b) Using techniques that eliminate the effects of these disturbance.
 - c) If external magnetic field and electrostatic field effect the reading, provide electrostatic & magnetic shields.



TYPES OF ERRORS

iii) Observational errors

- Introduced by the observer and are of the following types
 - a) Construction of the Scale : There is a possibility of error due to the division of the scale not being uniform and clear.
 - b) Fitness and Straightness of the Pointer : If the pointer is not fine and straight, then it always gives the error in the reading
 - c) Parallax : Without a mirror under the pointer there may be parallax error in reading
 - d) Efficiency or Skill of the Observer : Error in the reading is largely dependent upon the skill of the observer by which reading is noted accurately



TYPES OF ERRORS

➤ Random errors(Accidental errors)

- These are the errors that remain even after systematic errors have been substantially reduced.
- Generally an accumulation of a large number of small effects.
- Variable magnitude and sign & do not obey any known law.
- These errors may be either positive or negative .
- These are random error and their magnitudes are not constant.



TYPES OF ERRORS

- Persons performing the experiment have no control over the origin of these errors.
- Generally, these errors may be minimized by taking average of a large number of readings.

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CALIBRATION OF METERS

- Instrument calibration is one of the primary processes used to maintain instrument accuracy.
- Calibration is the process of configuring an instrument to provide a result for a sample within an acceptable range
- Eliminating or minimizing factors that cause inaccurate measurements is a fundamental aspect of instrumentation design
- So basically calibration of meters is the process of design of instruments by referencing standard instruments which are not used for ordinary purpose



INSTRUMENTS AND MEASUREMENT SYSTEMS



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HISTORY OF DEVELOPMENT OF INSTRUMENTS

➤ Mechanical Instruments

- The first instrument used by mankind were mechanical in nature
- Very reliable for static and stable condition
- Unable to respond rapidly to measurement of dynamic and transient condition.

➤ Electrical Instruments

- More rapid than mechanical instruments.
- But it depends upon a mechanical meter movement as indicating device.
- Hence these instruments have a limited time response.
- Eg:- energy meter



HISTORY OF DEVELOPMENT OF INSTRUMENTS

➤ Electronic Instruments

- Since the only movement require in electronic instrument is the movement of electrons, the response time is extremely small.
- Eg-CRO



CLASSIFICATION OF INSTRUMENTS

I) Broadly instruments are classified into two category.

1) **Absolute instrument**:- these instruments give the magnitude of quantity under measurement in terms of physical constants of the instrument.

Eg:- Tangent galvanometer.

2) **Secondary instruments**:- these instruments are calibrated by comparison with absolute instrument.

Eg:- galvanometer

Secondary instruments have scale and pointer display.



CLASSIFICATION OF INSTRUMENTS

II) Instruments can be classified into two major categories depending upon the way they present the result of measurement.

1) **Deflection type instrument**:- The deflection of pointer over a calibrated scale gives the quantity under measurement

Eg:-voltmeter

2) **Null type instrument**:- A null or zero indication leads to the determination of magnitude of unknown quantity

Eg:-potentiometer



CLASSIFICATION OF INSTRUMENTS

III) Secondary instruments can be divided into two

1) **Analog instruments**:- signals that vary in a continuous fashion and take on an infinite number of values in any given range are called analog signals and the instruments for measuring these signals are called analog instruments.

2) **Digital instruments**:- signals which vary in discrete steps and thus take up only finite different values in a given range are called digital signals. Instruments used for the measurement of digital signal are digital instruments



ANALOG INSTRUMENTS

CLASSIFICATION

I) Analog electrical instruments are classified into three according to the current that can be measured by them.

- 1) direct current instrument
- 2) alternating current instrument
- 3) both direct and alternating current instrument
(universal instrument)



ANALOG INSTRUMENTS

CLASSIFICATION

II) Depending upon their operation classified into three

1) Indicating instruments

- Indicate the value of a quantity being measured.
- Pointers moving over the scale give the indication

• E.g, Ammeters, Voltmeters , wattmeters etc.

2) Recording Instruments

- A continuous record of variations of the electrical quantity over a long period of time is given by these instruments
- It has a moving system which carries an inked pen which rests tightly on a graph chart
- E.g, Graphic recorders and galvanometer recorders are the examples of these instruments



ANALOG INSTRUMENTS

CLASSIFICATION

3) Integrating instrument:-

- The total amount of either electricity or electrical energy supplied over a period of time is measured by these instruments
- E.g:- Energy meters, Ampere hour meters etc



ANALOG INSTRUMENTS

CLASSIFICATION

III) Secondary analog instrument may be classified according to the principle of operation they utilize . The effects they utilize are,

- 1)magnetic effect
- 2)heating effect
- 3)electrostatic effect
- 4)electromagnetic effect
- 5)hall effect



INDICATING INSTRUMENTS OPERATING FORCES/TORQUES

- In order to ensure proper operation of indicating instruments, the following three torques are required
 - 1) Deflecting torque T_d or Operating torque
 - 2) Controlling torque T_c or Restoring torque
 - 3) Damping torque



DEFLECTING TORQUE (T_d)

- The deflecting torque is produced by utilizing the various effects of electric current or voltage (magnetic effect, induction effect, thermal effect, hall effect etc).
- This causes the moving system to move and hence the pointer to move from zero position.
- A deflecting system converts an electrical signal to a mechanical force.
- The method of producing this torque depends upon the type of instrument



CONTROLLING TORQUE (T_c)

- It is the torque which controls the movement of the pointer.
- The magnitude of movement of the moving system would be somewhat indefinite under the influence of deflecting torque, unless the controlling torque exerted to oppose the deflecting torque .
- It increases with increase in deflection of moving system
- Controlling torque is also called restoring or balancing torque



CONTROLLING TORQUE (T_c)

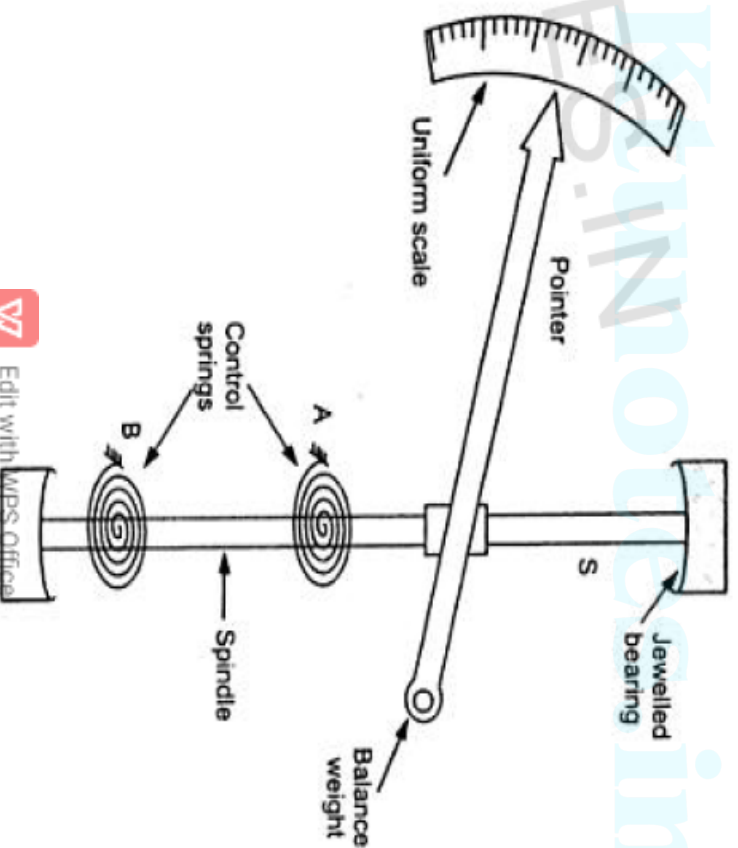
- The controlling torque serves two functions
 - The pointer stops moving beyond the final deflection
 - The pointer comes back to its zero position when the instrument is disconnected
- In indicating instruments, the controlling torque is obtained by two methods
 - i) Spring control
 - ii) Gravity control



CONTROLLING TORQUE (T_c)

i) Spring Control

- In the spring control method, a hair spring usually made of phosphor bronze, attached to the moving system in



CONTROLLING TORQUE (T_c)

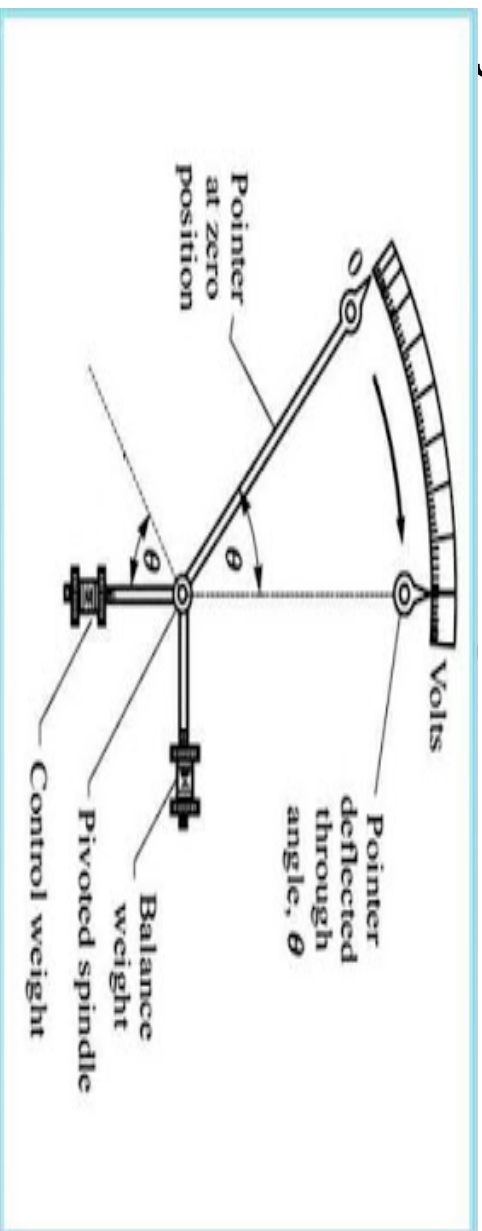
- With the deflection of the pointer, the spring is twisted in the opposite direction.
- This twist in the spring produces restoring torque which is directly proportional to the angle of deflection of the moving system.
- That is for spring control $T_c \propto \theta$
- The pointer comes to a position of rest (or equilibrium) when the deflecting torque (T_d) and the controlling torque (T_c) are equal .
- Instruments with spring control have a uniform scale over the entire range of measurement.



CONTROLLING TORQUE (T_c)

ii) Gravity Control

- In gravity control system, a small adjustable weight is placed on an arm attached to the moving system.
- The position of weight is adjustable.
- This weight produces a controlling torque due to gravity.



CONTROLLING TORQUE (T_c)

- When the pointer is at zero position as shown in fig 1 , the controlling torque is zero.

- When deflected by an angle θ , the weight exerts a force,

$$W \sin \theta$$

- The restraining or controlling torque is thus developed is given as

$$\tau_c = (W \sin \theta) \times L = WL \sin \theta$$

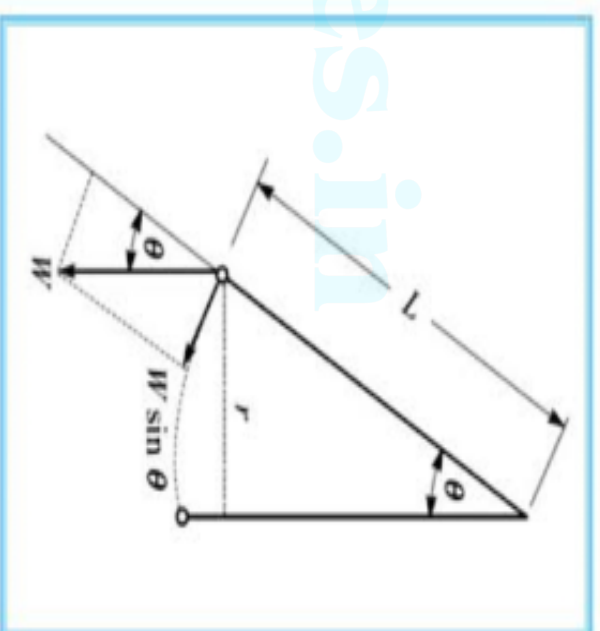
Since $\tau_d \propto I$, and $\tau_c = \tau_d$

$$\text{or } WL \sin \theta = kI$$

$$\Rightarrow I = \left(\frac{WL}{k} \right) \sin \theta$$

$$\text{or } I \propto \sin \theta$$

- The instruments employing gravity control must be in vertical position in order that the control may operate.



CONTROLLING TORQUE (T_c)

► Comparison of spring control and gravity control

	Gravity Control	Spring Control
1.	Adjustable small weight is used which produces the controlling torque.	Two hair springs are used which exert controlling torque.
2.	Controlling torque can be varied.	Controlling torque is fixed.
3.	The performance is not temperature dependent.	The performance is temperature dependent.
4.	The scale is nonuniform.	The scale is uniform.
5.	The controlling torque is proportional to $\sin \theta$.	The controlling torque is proportional to θ .
6.	The readings can not be taken accurately.	The readings can be taken very accurately.
7.	The system must be used in vertical position only.	The system need not be necessarily in vertical position.
8.	Proper levelling is required as gravity control.	The levelling is not required.
9.	Simple, cheap but delicate.	Simple, rigid but costlier compared to gravity control.
10.	Rarely used for indicating and portable instruments.	Very popularly used in most of the instruments.

DAMPING TORQUE

- Damping torque is one which acts on the moving system of the instrument only when it is moving and always opposes its motion.
- Such a stabilizing or damping torque is necessary to bring the pointer to rest quickly, otherwise due to inertia of the moving system the pointer will oscillate about its final deflected position for some time.



(a) The deflection of the pointer
From the scales Office



DAMPING TORQUE

- The function of damping is to absorb the energy from oscillating system and to bring it to rest in its equilibrium position
- The damping force can be produced by
 - i) Air friction damping
 - ii) Fluid friction damping
 - iii) Eddy current damping



DAMPING TORQUE

i) Air friction damping

- Two arrangements of air friction damping are shown in fig.

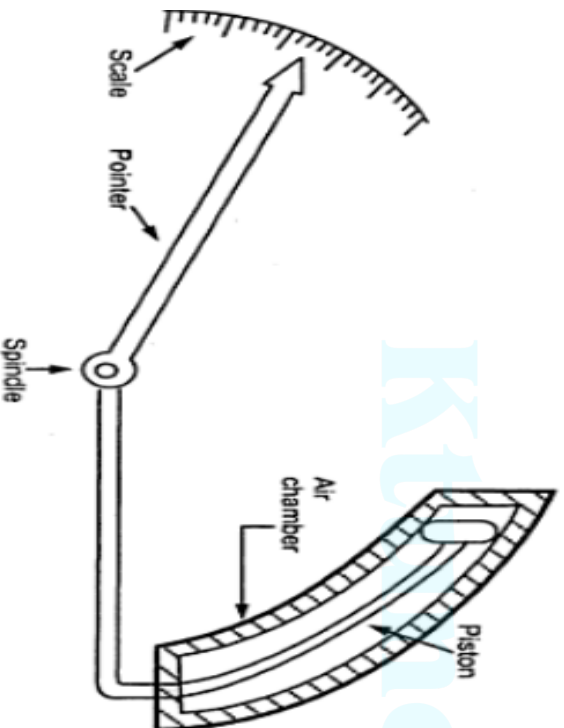


Fig 1

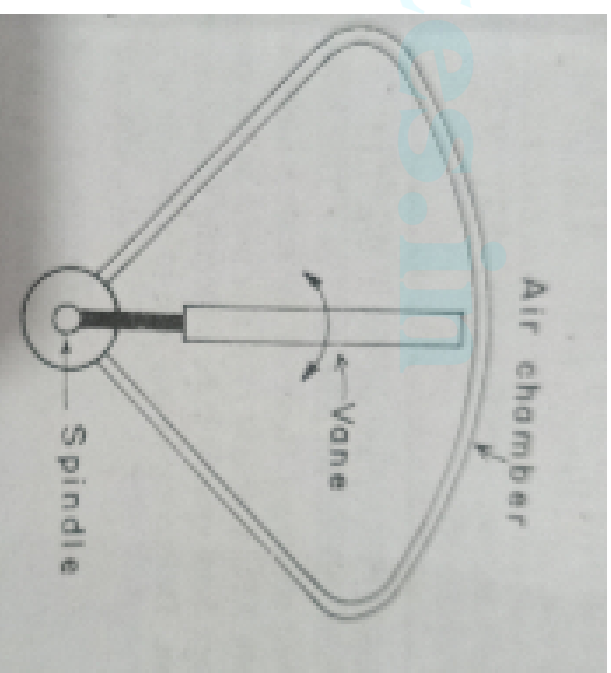


Fig 2



DAMPING TORQUE

- In the first arrangement (fig 1) , a light aluminium piston is attached to the spindle that carries the pointer and moves with a very little clearance in a rectangular or circular air chamber closed at one end
- The cushioning action of the air on the piston damps out any tendency of the pointer to oscillate about the final deflected position .



DAMPING TORQUE

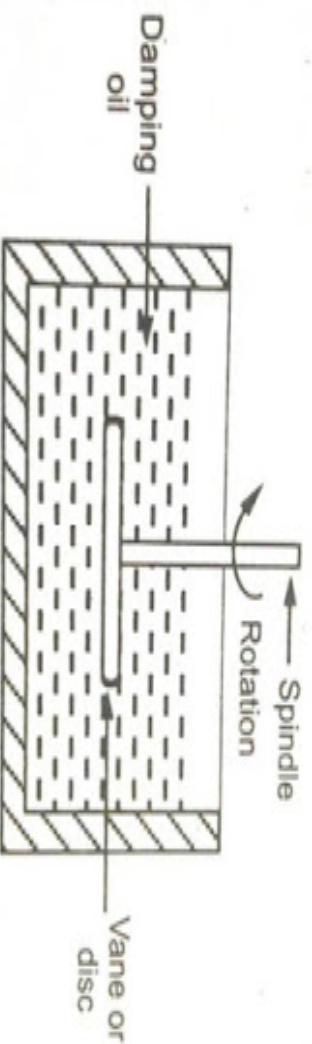
- In the second arrangement (fig 2) one or two light aluminium vanes are attached to the same spindle that carries the pointer.
- The vanes are permitted to swing in a sector shaped closed box that is just large enough to accommodate the vanes.
- As the pointer moves, the vanes swing in a box compressing the air in front of them.
- The pressure of compressed air in the vanes provides the necessary damping force to reduce the tendency of the pointer to oscillate .



DAMPING TORQUE

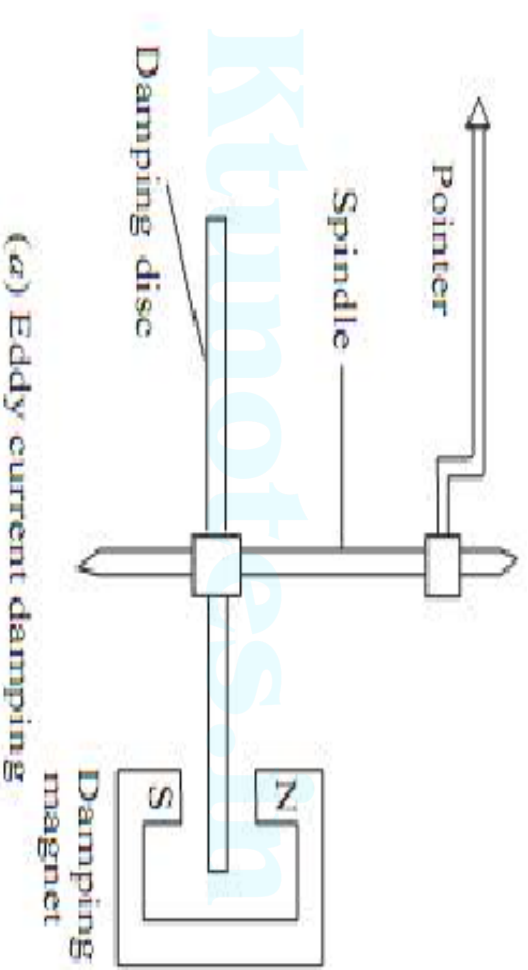
ii) Fluid friction damping

- In this method discs or vanes attached to the spindle of the moving system are kept immersed in a pot containing oil of high viscosity
- As the pointer moves, the friction between the oil and vanes opposes the motion of the pointer and thus necessary damping is provided



DAMPING TORQUE

iii) Eddy current damping



- In Eddy current damping, a thin aluminium or copper disc attached to the moving system is allowed to pass between the poles of a permanent magnet.



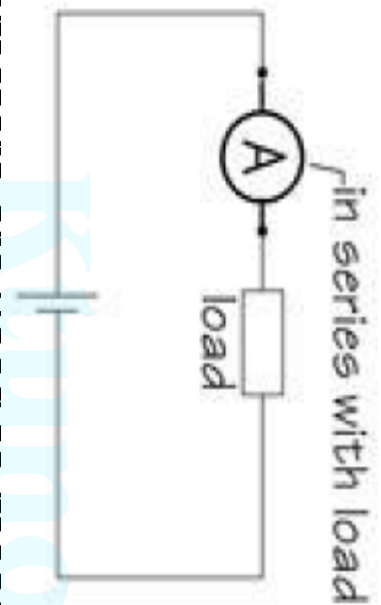
DAMPING TORQUE

- As the pointer moves, the disc cuts the magnetic field and eddy currents are induced in the disc.
- These eddy currents react with the field of the magnet to produce a force which opposes the motion(Lenz's law).
- In this way eddy currents provide the damping torque to reduce the oscillations of the pointer .



AMMETERS AND VOLTMETERS

► Ammeters

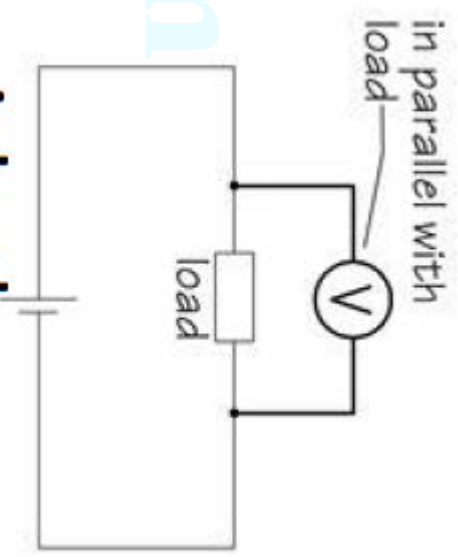


- Ammeters are used to measure current in amperes and are connected in series with the circuit whose current is to be measured.
- The power loss in an ammeter is IR where I is the current to be measured and R is the resistance of the ammeter.
- Ammeters have low electrical resistance so that they cause small voltage drop and consequently absorb small power.

AMMETERS AND VOLTMETERS

➤ Voltmeter

- Voltmeters are used to measure voltage in volts and are connected in parallel with the circuit whose voltage is to be measured
- The power loss in a voltmeter is (V^2/R_v) , where V is the voltage to be measured and R_v is the resistance of the voltmeter.
- Therefore voltmeters have a very high electrical resistance, in order that the current drawn by them is small and consequently the power consumed is small.



TYPES OF INSTRUMENTS

- The main types of instruments used as ammeters and voltmeters are

- i) Permanent magnet moving coil (PMMC)-dc only
- ii) Moving iron- both ac & dc
- iii) Electro-dynamometer- both ac & dc
- iv) Hot wire- both ac & dc
- v) Thermocouple- both ac & dc
- vi) Induction-ac only
- vii) Electrostatic- both ac & dc
- viii) Rectifier - both ac & dc



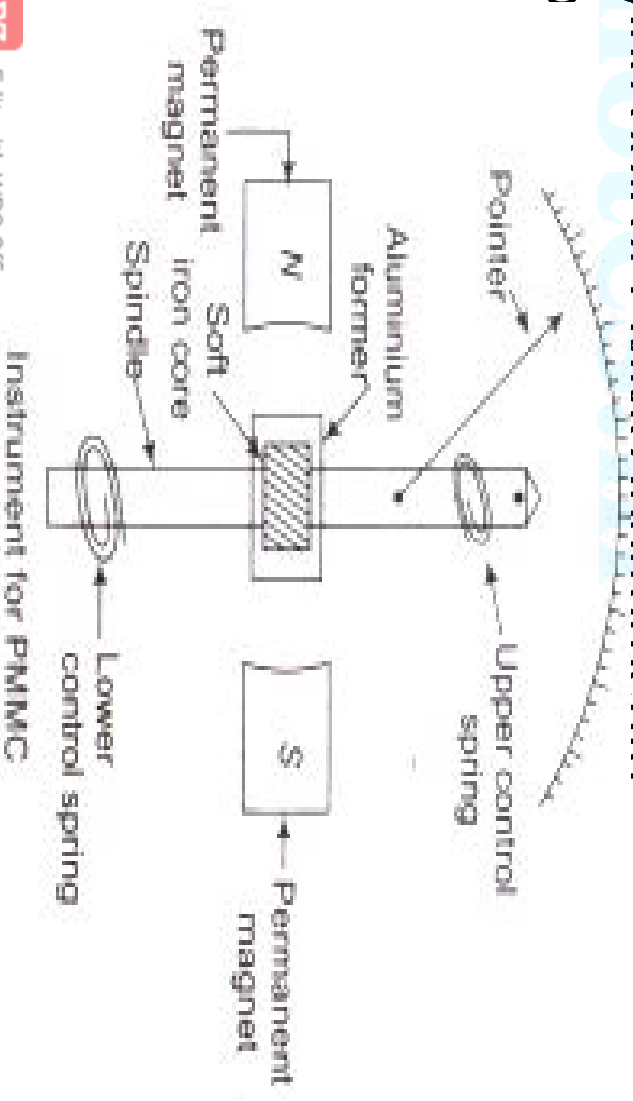
PERMANENT MAGNET MOVING COIL INSTRUMENT (PMMC)

- Permanent Magnet Moving Coil or PMMC Instruments are the most accurate type for the measurement of DC current or voltage.

► Construction of PMMC Instruments

- Permanent Magnet Moving Coil or PMMC Instruments consists of following components

- Deflecting system
- Magnet System
- Control Spring
- Damping
- Pointer and Scale



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PERMANENT MAGNET MOVING COIL INSTRUMENT (PMMC)

➤Deflecting system:

- The moving coil made up of copper is wound with many turns on the rectangular Aluminum former.
- This Aluminum former is pivoted on the jewelled bearing.
- The coil can move freely in the magnetic field produced by the Permanent Magnet System.
- The current or voltage to be measured is passed through the coil.
- The coil experience a force and provide necessary deflection.



PERMANENT MAGNET MOVING COIL INSTRUMENT (PMMC)

➤ Magnet System

- Simple U shaped permanent magnet made of Alcomax or Alnico is widely use in PMMC instruments.
- These magnets have high coercive force and can produce field of the order of 0.1 to 1 Wb/m².

➤ Control Spring

- The controlling torque in PMMC Instruments is provided by two phospher bronze hair springs mounted on the jewel bearing.
- The springs also serve to lead current in and out of the coil.



PERMANENT MAGNET MOVING COIL INSTRUMENT (PMMC)

➤Damping:

- Damping torque in PMMC instruments are produced by the movement of Aluminum former in the magnetic field of Permanent Magnet.
- Due to movement of Aluminum former an emf is induced resulting in eddy current which opposes the motion thereby provides a damping torque.

➤Pointer and Scale

- The pointer is carried by the spindle and moves over a graduated scale.



PERMANENT MAGNET MOVING COIL INSTRUMENT (PMMC)

➤ Principle of operation

- When D.C. supply is given to the moving coil, D.C. current flows through it.
- When the current carrying coil is kept in the magnetic field, it experiences a force. This force produces a torque and the former rotates.
- The pointer is attached with the spindle. When the former rotates, the pointer moves over the calibrated scale.
- When the polarity is reversed a torque is produced in the opposite direction .



PERMANENT MAGNET MOVING COIL INSTRUMENT (PMMC)

- The mechanical stopper does not allow the deflection in the opposite direction.
- The polarity should be maintained with PMMC instrument.
- If A.C. is supplied, a reversing torque is produced. This cannot produce a continuous deflection. Therefore this instrument cannot be used in A.C .



PERMANENT MAGNET MOVING COIL INSTRUMENT (PMMC)

►Torque developed

- N – Number of turns of coil
- l, b – the vertical and horizontal length of the side.
- I – current through the coil.
- B – flux density in the air gap

Force produced in the coil $F = NBI I \sin \theta$

Field is radial so $\theta = 90$

$$F = NBI I$$

$$T_d = \text{Force} \times \text{Distance}$$

$$T_d = NBI I \times b = NBI A$$

$$T_d \propto I \quad \text{ie} \quad T_d = GI$$

PERMANENT MAGNET MOVING COIL INSTRUMENT (PMMC)

$$T_c = K\theta$$

Where K = Spring constant

θ = Angular movement of coil.

– At steady state condition, deflecting and controlling torque shall be equal,

$$T_d = T_c \quad \Rightarrow \quad GI = K\theta$$

$$\Rightarrow \theta = (G / K)I$$

– Deflection directly proportional to I Passing through the meter scale is uniform (linear)



PERMANENT MAGNET MOVING COIL INSTRUMENT (PMMC)

► Advantages of permanent magnet moving coil instruments

1. The scale is uniformly divided as the current is directly proportional to deflection of the pointer. Hence it is very easy to measure quantities from these instruments.
2. Power consumption is also very low in these types of instruments.
3. A high torque to weight ratio.
4. These are having multiple advantages; a single instrument can be used for measuring various quantities by using different values of shunts and multipliers.
5. Damping effective and reliable.
6. No hysteresis loss because of AI former.



PERMANENT MAGNET MOVING COIL INSTRUMENT (PMMC)

➤ Disadvantages of permanent magnet moving coil instruments

1. These instruments cannot measure AC quantities.
2. The cost of these instruments is high as compared to moving iron instruments.
3. Error due to ageing of magnets.



PERMANENT MAGNET MOVING COIL INSTRUMENT (PMMC)

➤ Errors in PMMC instruments

- Frictional error.
- Temperature error: As temp increases spring becomes more flexible and magnets weaker.
- Error owing to weakening of magnet due to ageing.
- Stray magnetic field: usually not affected but due to presence of iron in working parts effects of external magnetic field increases.



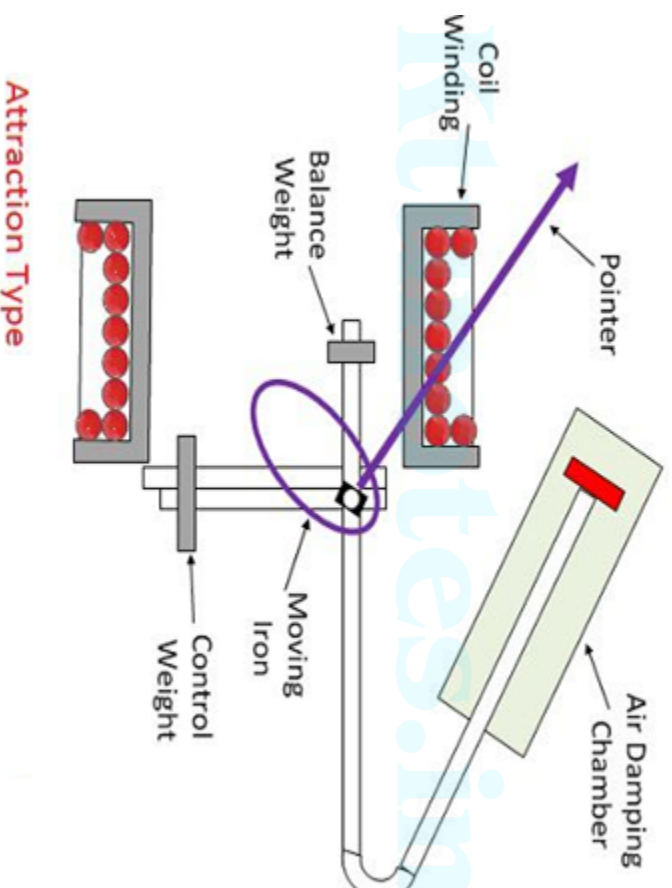
MOVING IRON INSTRUMENTS (MI)

- This type of instrument is principally used for the measurement of alternating currents and voltages.
- This can also be used for DC measurements.
- There are two types of moving iron instruments.
 - 1) Attraction type
 - 2) Repulsion type



MOVING IRON INSTRUMENTS (MI)

► Attraction type MI instrument



MOVING IRON INSTRUMENTS (MI)

- It consist of a flat coil which has a narrow slot like opening.
- The moving iron is a flat disc or sector eccentrically pivoted just outside the coil.
- A pointer is attached to the spindle so that it is deflected with the motion of the soft iron piece.
- When current passes through the coil ,a magnetic field is developed.
- The moving iron piece moves from a position of weaker field outside to a position of stronger field inside.
- Or in other words the moving iron is attracted in.



MOVING IRON INSTRUMENTS (MI)

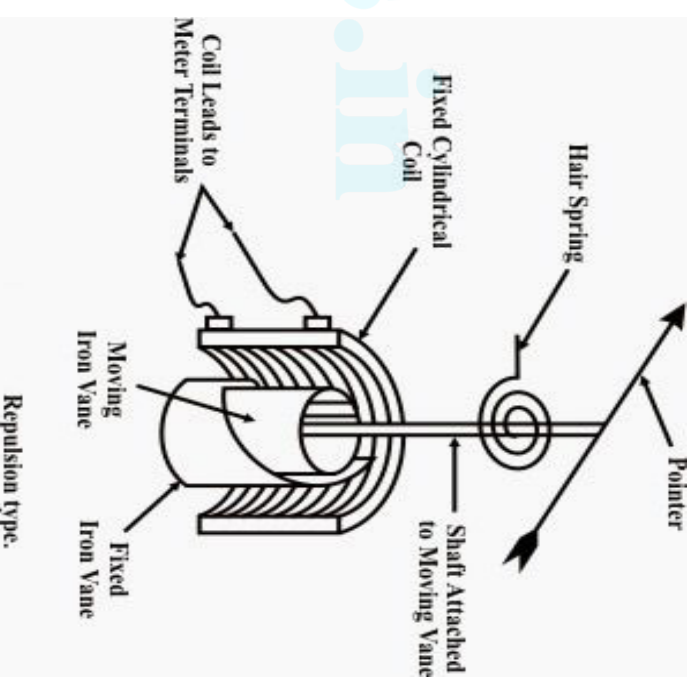
- The controlling torque is produced by springs but gravity control can be used for panel type of instruments which are vertically mounted.
- A piston, attached to the spindle, moves inside an air chamber and gives the necessary damping.
- If current in the coil is reversed, the direction of magnetic field also reverses and so does the magnetism produced in the soft iron piece.
- Hence the direction of the deflecting torque remains unchanged
- For this reason , such instruments can be used for both ac and dc measurements .



MOVING IRON INSTRUMENTS (MI)

► Repulsion Type MI Instrument

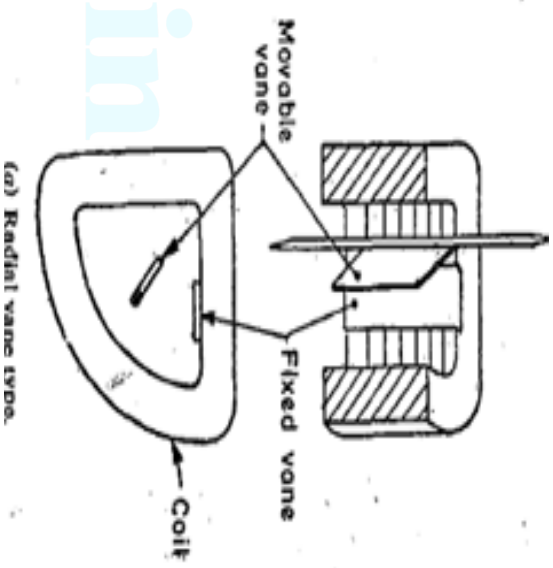
- In the repulsion type, there are two vanes inside the coil, one fixed and other movable.
- These are similarly magnetized when the current flows through the coil and there is a force of repulsion between the two vanes resulting in the movement of the moving vane.
- The movable vane is attached to the spindle which carries the pointer will move away from the fixed vane resulting in deflection of the pointer.
- Controlling torque is provided by spring or gravity.
- Air friction damping is provided.



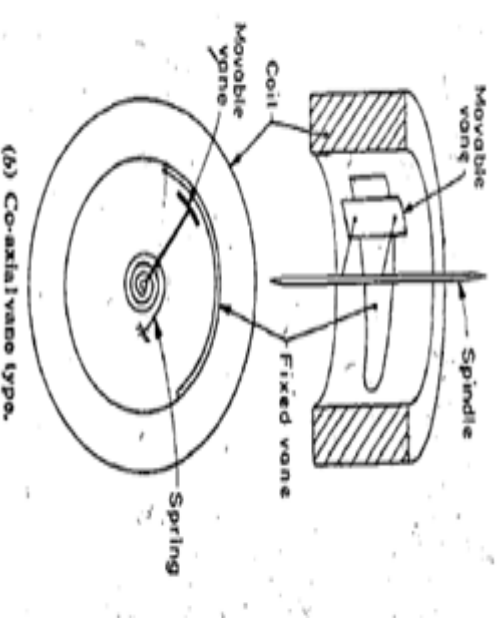
MOVING IRON INSTRUMENTS (MI)

- Two different designs are in common use ,

(i) Radial Vane Type:- In this type, the vanes are radial strips of iron. The strips are placed within the coil . The fixed vane is attached to the coil and the movable one to the spindle of the instrument.



(ii) Co-axial Vane Type:- In this type of instrument, the fixed and moving vanes are sections of co-axial cylinders. The fixed and moving vanes .



MOVING IRON INSTRUMENTS (MI)

► Torque equation of MI instruments

- The expression for torque of a moving iron instrument can be derived by considering the energy relations when there is a small increment in current supplied to the instrument.
- This result in a small deflection $d\theta$ and some mechanical work will be done.
- Let Td be the deflecting torque.
- Therefore mechanical work done = torque $\times$ angular displacement

$$= T_d \cdot d\theta$$

- Due to the change in inductance there will be a change in the energy stored in the magnetic field.



MOVING IRON INSTRUMENTS (MI)

- Let I be the initial current, L be the instrument inductance and θ is the deflection. If the current increases by dI then it causes the change in deflection $d\theta$ and the inductance by dL .
- In order to involve the increment dI in the current, the applied voltage must be increase by:

$$e = \frac{d\phi}{dt} = \frac{d}{dt}(LI) = I \frac{dL}{dt} + L \frac{dI}{dt}$$

The electrical energy supplied is $eIdt = I^2 dL + LI dI$



MOVING IRON INSTRUMENTS (MI)

The current is changes from I to $(I + dI)$, and the inductor L to $(L + dL)$

Therefore the stored energy changes from $= \frac{1}{2} I^2 L$ to $\frac{1}{2} (I + dI)^2 (L + dL)$

Hence the change in stored energy $= \frac{1}{2} (I + dI)^2 (L + dL) - \frac{1}{2} I^2 L$

As dI and dL are very small, neglecting the second and higher order terms in small quantities, this

becomes $LdI + \frac{1}{2} I^2 dL$



MOVING IRON INSTRUMENTS (MI)

From the principle of conservation of energy,

Electrical energy supplied = Increase in stored energy + Mechanical work done

$$I^2 dL + ILdl = ILdl + \frac{1}{2} I^2 dL + T_d d\theta$$

$$\therefore T_d d\theta = \frac{1}{2} I^2 dL$$

$$\text{or deflecting torque } T_d = \frac{1}{2} I^2 \frac{dL}{d\theta}$$



MOVING IRON INSTRUMENTS (MI)

where T_d is in newton-metre, I is in ampere, L is in henry and θ is in radians.

The moving system is provided with control springs and in turn the deflecting torque T_d is balanced by the controlling torque $T_c = k \theta$

where k is the control spring constant (N-m/rad) and θ is the deflection in radians.

At final steady position, $T_c = T_d$

$$\text{or} \quad k\theta = \frac{1}{2} I^2 \frac{dL}{d\theta}$$

$$\therefore \text{deflection } \theta = \frac{1}{2} \frac{I^2}{k} \frac{dL}{d\theta}$$

Hence, the deflection is proportional to square of the rms value of the operating current. The deflection torque is, therefore, unidirectional whatever may be the polarity of the current.

Scale is nonuniform



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MOVING IRON INSTRUMENTS (MI)

► Advantages of MI instruments

- Universal use-can be used for both ac and dc
- Less frictional error-since current carrying part (heavy part) is stationary and the moving parts are light in weight.
- Cheapness
- Robustness
- Reasonably accurate

► Disadvantages of MI instruments

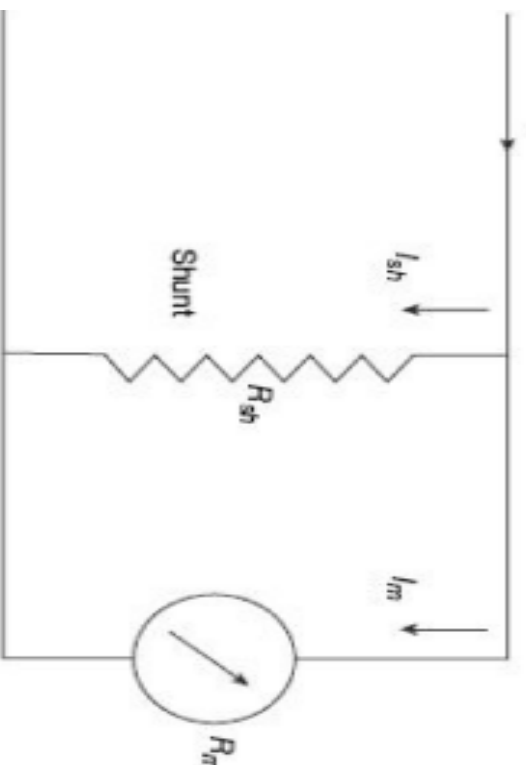
- Have non uniform scale
- Not as sensitive as PMMC instruments



EXTENSION OF RANGE OF METERS- (PMMC)

► Extension of range of ammeter

- Shunts:-A low resistance called as shunt is connected in parallel with the ammeter to extent the range of current measurement.



EXTENSION OF RANGE OF METERS- (PMMC)

The resistance of the shunt can be calculated using conventional circuit analysis.

R_{sh} = shunt resistance (Ω)

R_m = coil resistance (Ω)

$I_m = I_{fs}$ = full-scale deflection current (A)

I_{sh} = shunt current (A)

I = current to be measured (A)

$$I_{sh}R_{sh} = I_m R_m$$

$$I = I_{sh} + I_m$$

$$I_{sh} = I - I_m$$

$$R_{sh} = \frac{I_m}{I_{sh}} R_m$$

$$R_{sh} = \frac{I_m}{I - I_m} R_m$$



EXTENSION OF RANGE OF METERS- (PMMC)

The ratio of the total current to the current in the meter is called *multiplying power of shunt*. Multiplying power,

$$m = \frac{I}{I_m} = 1 + \frac{R_m}{R_{sh}}$$

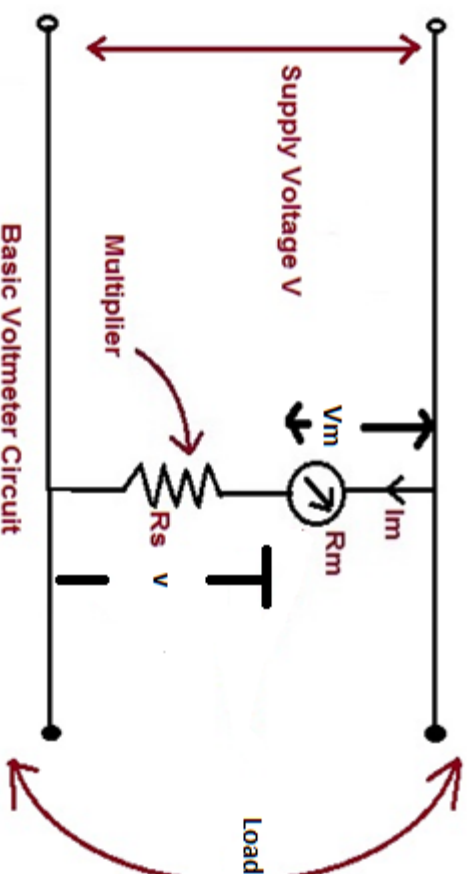
$$\therefore R_{sh} = \frac{R_m}{m-1}$$

- The general requirements of shunts are
 - It should carry current without excessive temperature rise
 - The change in resistance due to change in temperature is negligible
 - The resistance of shunt should not vary with time
- Shunt resistance is usually made of manganin.



EXTENSION OF RANGE OF METERS- (PMMC)

- Voltmeter multipliers
- A large resistance known as multiplier is connected in series with the voltmeter to extent the range of voltage measurement.



EXTENSION OF RANGE OF METERS- (PMMC)

The value of multiplier required to extend the voltage range, is calculated as under:

R_s = multiplier resistance (Ω)

R_m = meter resistance (Ω)

$I_m = I_{fs}$ = full scale deflection current (A)

V_m = voltage across the meter for producing current I_m (A)

V = voltage to be measured (A)

$$\frac{V}{R_s} = \frac{V_m}{R_m}$$

$$\frac{V - V_m}{R_s} = \frac{V_m}{R_m}$$

$$R_s = \frac{(V - V_m)R_m}{V_m}$$

$$R_s = \left(\frac{V}{V_m} - 1\right)R_m$$

$m = V/V_m$ is the multiplication factor

$$R_s = (m - 1)R_m$$



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EXTENSION OF RANGE OF METERS- (PMMC)

- The general requirements of multiplier are
 - The change in resistance due to change in temperature is negligible.
 - The resistance of multiplier should not vary with time.
- Multiplier resistance is usually made of manganin and constantan.



PROBLEMS

Example 1:- A moving coil ammeter has a full scale deflection of $50 \mu\text{A}$ and a coil resistance of 1000Ω . What will be the value of the shunt resistance required for the instrument to be converted to read a full scale reading of 1 Amp.

Solution 1:- Full scale deflection current $I_m = 50 * 10^{-6} \text{ A}$

Instrument resistance $R_m = 1000 \Omega$

Total current to be measured $I = 1 \text{ A}$

Resistance of ammeter shunt required $R_{sh} = \frac{R_m}{\frac{I}{I_m} - 1} = \frac{1000}{\frac{1}{50 * 10^{-6}} - 1}$



PROBLEMS

Example 2:- The full scale deflection current of an ammeter is 1 mA and its internal resistance is 100 Ω . If this meter is to have scale deflection at 5 A, what is the value of shunt resistance to be used.

Solution 2:- Full scale deflection current $I_m = 1 \text{ mA} = 0.001 \text{ A}$
Instrument resistance $R_m = 100 \text{ } \Omega$.

Total current to be measured $I = 5 \text{ A}$

Resistance of ammeter shunt required $R_{sh} = \frac{R_m}{\frac{I}{I_m} - 1} = \frac{100}{\frac{5}{0.001} - 1}$

$$R_{sh} = 0.020004 \text{ } \Omega$$



PROBLEMS

Example 3:- The full scale deflection current of a meter is 1 mA and its internal resistance is 100 Ω . If this meter is to have full-scale deflection when 100 V is measured. What should be the value of series resistance?

Solution 3:- Instrument resistance $R_m = 100 \Omega$

Full-scale deflection current $I_m = 1 \text{ mA} = 1 * 10^{-3} \text{ A}$

Voltage to be measured $V = 100 \text{ V}$

$$\text{Required series resistance } R_{se} = \frac{V}{I_m} - R_m = \frac{100}{1 * 10^{-3}} - 100 = 99,900 \Omega$$

